

Primes in Legendre intervals over $\mathbb{F}_q[t]$: an exact explicit-formula algorithm, verification for $d \leq 10$, and an anomaly in characteristic 2

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Abstract

For monic $f \in \mathbb{F}_q[t]$ of degree d , the Legendre interval $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$ is the analogue of $[n^2, (n+1)^2]$. The Hayes–Weil estimate shows that $\mathcal{I}_f$ contains an irreducible polynomial whenever $q \geq d-1$; the range $q \leq d-2$ is open. We give an algorithm that computes the prime count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ for *all* q^{d-1} intervals of a given (q, d) at once. It evaluates every Hayes L -function by a fast transform over the Hayes group, whose structure we make explicit for every $\mathbb{F}_q$, and runs in $O(d^2 q^{d-1} + d \sigma q^{d-1})$ operations, where σ is the sum of the cyclic orders of the group. Carried out in exact modular arithmetic, it proves that every Legendre interval contains an irreducible polynomial for every prime power q when $d \leq 10$, and for $q = 2$ with $d \leq 25$, $q = 3$ with $d \leq 16$, $q = 4$ with $d \leq 13$ and $q = 5$ with $d \leq 11$. For odd q the fluctuations of Ψ across intervals follow the Keating–Rudnick variance $(d-2)q^{d+1}$ and a Gaussian law already for $q = 3$, and the form factor of the L -functions matches the circular unitary ensemble up to an explicit arithmetic diagonal term. In characteristic 2 the Legendre intervals form the subgroup of squares, every element has the shape $h^2 + tb^2$, and the variance of Ψ over them differs from the full-group value, which does follow $d-2$, by factors between 0.33 and 21. We show that this discrepancy is carried entirely by correlations between L -functions twisted by quadratic characters, and state its explanation as an open problem.

Contents

1	Introduction	1
2	Setup	2
3	The structure of the Hayes group	2
4	The algorithm	3
5	Verification	3
6	Fluctuations across intervals	4
7	The zeros	4

8	Characteristic 2	6
8.1	Structure	6
8.2	The anomaly	6
9	Discussion	7

1 Introduction

Legendre’s conjecture asserts that there is a prime between consecutive squares. Its function field analogue concerns the *Legendre interval*

$$\mathcal{I}_f = \{f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d\}, \quad f \in \mathbb{F}_q[t] \text{ monic of degree } d,$$

the set of monic polynomials of degree $2d$ whose coefficients of $t^{2d-1}, \dots, t^{d+1}$ agree with those of f^2 . Write

$$\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P), \quad N_{\text{irr}}(f) = \#\{P \in \mathcal{I}_f : P \text{ irreducible}\},$$

so that $2d N_{\text{irr}}(f) = \Psi(f) - E(f)$ with $E(f)$ the contribution of proper prime powers.

What is known. Hayes characters and Weil’s theorem give $|\Psi(f) - q^{d+1}| \leq (d-2)(q^d - q)$ in every characteristic [5, 6, 3, 4]; with a bound for $E(f)$ this shows $N_{\text{irr}}(f) \geq 1$ whenever $q \geq d-1$ [2]. At $q = d-2$ the main term and the error are of the same size, and for $q \leq d-2$, which contains the genuine analogue (q fixed, $d \rightarrow \infty$), no method is known. Earlier computations [2] tested every element of every interval and reached $d \leq 8$ for all q .

This paper.

- (1) *An exact all-intervals algorithm* (Sections 3–4). The values $\Psi(f)$ for all q^{d-1} classes are obtained from the explicit formula, computing every Hayes L -function by a transform over the Hayes group. We determine an explicit basis of this group for every $\mathbb{F}_q$. Done modulo two primes and recombined, the computation is exact.
- (2) *Verification* (Section 5). Every Legendre interval contains an irreducible polynomial for all q when $d \leq 10$, and much further for small q .
- (3) *Fluctuations and zeros* (Sections 6–7). For odd q , the variance across intervals agrees with Keating–Rudnick [7] at fixed q ; the form factor of the L -functions equals an exact arithmetic diagonal for $n \leq d-1$ and the circular unitary ensemble (CUE) value beyond.
- (4) *Characteristic 2* (Section 8). The Legendre intervals are the squares of the Hayes group; their prime counts fluctuate far more than predicted. We locate the excess in quadratic twists and pose its explanation as an open problem.

2 Setup

Let $\ell = d-1$. For monic g of degree k put $[g] = u^k g(1/u) \bmod u^{\ell+1}$, an element of

$$\mathcal{G} = \mathcal{G}_\ell = \{1 + c_1 u + \dots + c_\ell u^\ell\} \quad (\text{multiplication mod } u^{\ell+1}), \quad |\mathcal{G}| = q^\ell.$$

The map $g \mapsto [g]$ is multiplicative and records the ℓ coefficients below the leading one. For $x \in \mathcal{G}$ let $\Psi(x) = \sum_{\deg P=2d, [P]=x} \Lambda(P)$; then $\Psi(f) = \Psi([f^2]) = \Psi([f]^2)$.

Lemma 2.1. *As f runs over monic polynomials of degree d , $[f]$ runs over all of $\mathcal{G}$. Hence the Legendre intervals are exactly the classes in $\mathcal{G}^2 = \{x^2 : x \in \mathcal{G}\}$. If q is odd, $\mathcal{G}^2 = \mathcal{G}$.*

Proof. $[f]$ is determined by, and determines, the coefficients $f_{d-1}, \dots, f_1$. Squaring is a bijection on the finite p -group $\mathcal{G}$ when p is odd. $\square$

For a character χ of $\mathcal{G}$ and $n \geq 1$ put $\psi_n(\chi) = \sum_{\deg g=n} \Lambda(g)\chi([g])$ and $c_k(\chi) = \sum_{\deg g=k} \chi([g])$. For $\chi \neq 1$, $L(z, \chi) = \sum_k c_k(\chi)z^k$ is a polynomial of degree $\leq \ell - 1$ whose inverse roots have absolute value $q^{1/2}$ or 1 [5, 7], and $zL'(z, \chi)/L(z, \chi) = \sum_n \psi_n(\chi)z^n$. Comparing coefficients gives Newton's recursion

$$\psi_n(\chi) = n c_n(\chi) - \sum_{k=1}^{n-1} \psi_k(\chi) c_{n-k}(\chi), \quad c_n = 0 \ (n \geq \ell), \quad (1)$$

which involves no division. By orthogonality, for every $x \in \mathcal{G}$,

$$\Psi(x) = q^{-\ell} \sum_{\chi} \overline{\chi(x)} \psi_{2d}(\chi), \quad \psi_{2d}(1) = q^{2d}. \quad (2)$$

3 The structure of the Hayes group

Let $q = p^k$ and let $\omega_0, \dots, \omega_{k-1}$ be a basis of $\mathbb{F}_q$ over $\mathbb{F}_p$. For $j \geq 1$ with $p \nmid j$ let s_j be the least integer with $jp^{s_j} > \ell$.

Proposition 3.1. $\mathcal{G}_\ell$ is the internal direct product of the cyclic groups generated by $1 + \omega_i u^j$, $p \nmid j \leq \ell$, $0 \leq i < k$, and $1 + \omega_i u^j$ has order p^{s_j} . In particular $\mathcal{G}_\ell \cong \prod_{p \nmid j \leq \ell} (\mathbb{Z}/p^{s_j})^k$.

Proof. In characteristic p , $(1 + \omega u^j)^{p^s} = 1 + \omega^{p^s} u^{jp^s}$, which is 1 modulo $u^{\ell+1}$ exactly when $jp^s > \ell$; this gives the orders. The orders multiply to $\prod_j q^{s_j} = q^{\#\{(j,s): jp^s \leq \ell\}} = q^\ell$, because every $m \leq \ell$ is uniquely jp^s with $p \nmid j$. It remains to show that the elements generate. Let $x = 1 + c_m u^m + O(u^{m+1})$ with $c_m \neq 0$, write $m = jp^a$, and expand $c_m^{-a} = \sum_i e_i \omega_i$ with $e_i \in \mathbb{F}_p$. Then $\prod_i (1 + \omega_i u^j)^{e_i p^a} = 1 + \sum_i e_i \omega_i^{p^a} u^m + O(u^{m+1}) = 1 + c_m u^m + O(u^{m+1})$, so dividing x by this product raises the order of vanishing of $x - 1$. Induction on m expresses x as a product of the proposed generators. $\square$

The proof is an algorithm: it computes the coordinates of $[g]$ (its *discrete logarithm*) in $O(\ell^2 p)$ operations in $\mathbb{F}_q$. Characters of $\mathcal{G}$ are then $\chi_b(x) = \prod_{(j,i)} \zeta_{p^{s_j}}^{-a_{j,i}(x) b_{j,i}}$, and sums over $\mathcal{G}$ weighted by characters become multidimensional discrete Fourier transforms with axis lengths p^{s_j} .

4 The algorithm

1. For $k = 0, \dots, \ell - 1$ form the indicator of $\{[g] : \deg g = k\}$ in the coordinates of Proposition 3.1 (for $k \leq \ell$ distinct g give distinct classes), and transform it to obtain $c_k(\chi)$ for all χ .
2. For every χ run (1) up to $n = 2d$.
3. Set $\psi_{2d}(1) = q^{2d}$ and invert the transform to obtain $\Psi(x)$ for all x by (2).

Step 1 costs ℓ transforms, step 2 costs $O(d^2)$ per character, step 3 one transform. With naive transforms along each axis the total is $O(d^2 q^\ell + d \sigma q^\ell)$ operations, where $\sigma = k \sum_{p \nmid j \leq \ell} p^{s_j}$. Testing every polynomial costs about q^{2d} instead.

Exact arithmetic. All quantities in (1)–(2) lie in $\mathbb{Z}[\zeta][1/q]$, where ζ is a primitive p^S -th root of unity and $p^S = \max_j p^{s_j}$. For a prime $M \equiv 1 \pmod{p^S}$, $M \neq p$, choose $z \in \mathbb{F}_M$ of order p^S ; the ring map $\zeta \mapsto z$ turns (1)–(2) into identities in $\mathbb{F}_M$. Running the algorithm in $\mathbb{F}_M$ therefore returns $\Psi(x) \bmod M$ exactly. Since $0 \leq \Psi(x) \leq 2d q^{d+1}$, two primes $M_1, M_2 < 2^{31}$ with $M_1 M_2 > 2d q^{d+1}$ determine $\Psi(x)$ by the Chinese remainder theorem. All products fit in 64-bit integers.

Checks. A floating-point version (fast Fourier transforms) is used for exploration. The exact and floating-point results agree on every class tested; both agree with brute-force counts for all 11,232 orbit representatives computed in [2]; and $\sum_x \Psi(x) = q^{2d}$ holds exactly in every run. For example $q = 5$, $d = 8$ took one hour by brute force in [2] and 0.4 s here.

5 Verification

An interval is *certified* if $\Psi(f) > d q^a + 2q^{\lfloor 2d/3 \rfloor}$, with $a = 1$ for odd q and $a = \lceil (d+1)/2 \rceil$ for even q ; by [2, Lemma 2.3] this bound exceeds $E(f)$, so a certified interval contains an irreducible polynomial.

Theorem 5.1 (computer-assisted). *Every Legendre interval over $\mathbb{F}_q$ contains an irreducible polynomial in each of the following cases: $q = 2$, $d \leq 25$; $q = 3$, $d \leq 16$; $q = 4$, $d \leq 13$; $q = 5$, $d \leq 11$; $q = 7$, $d \leq 10$; $q = 8$, $d \leq 10$; $q = 9$, $d \leq 8$. Consequently the function field analogue of Legendre’s conjecture holds for every prime power q when $d \leq 10$.*

Proof. For each listed (q, d) the exact algorithm computes Ψ on every Legendre class; all classes are certified except the cases $(2, 2)$, $(2, 3)$, $(2, 4)$, $(2, 6)$, whose exact counts are positive [2]. For the second statement let $d \leq 10$: if $q \geq d - 1$ use [2]; otherwise $q \leq d - 2 \leq 10 - 2$ and q is one of the listed prime powers. $\square$

The case $d = 11$ would require $q = 9$ with $9^{10} \approx 3.5 \times 10^9$ classes, beyond the memory used here.

6 Fluctuations across intervals

For odd q every class is a Legendre interval (Lemma 2.1), so the distribution of Ψ over Legendre intervals is the distribution over all q^{d-1} classes. Keating and Rudnick [7] proved $\text{Var } \Psi \sim (d-2)q^{d+1}$ as $q \rightarrow \infty$ (for $d \geq 4$).

For odd q the ratio $\text{Var } \Psi / ((d-2)q^{d+1})$ is between 0.96 and 0.99 for the largest computed d of each q (Figure 1), and Z is close to Gaussian (Figure 2). The minimum of Ψ approaches q^{d+1} as d grows, as the Gaussian heuristic of [2] predicts.

7 The zeros

For each nontrivial χ the inverse roots of $L(z, \chi)$ were computed numerically. In all cases listed in Table 2 every inverse root has absolute value $\sqrt{q}$ to within 10^{-4} , and a proportion close to $1 - 1/q$ of the characters has exactly $\ell - 1 = d - 2$ of them.

Let $F(n) = (q^\ell - 1)^{-1} \sum_{\chi \neq 1} |\psi_n(\chi)|^2 / q^n$. Writing the inverse roots as $\sqrt{q} e^{i\theta}$, $F(n)$ is the average of $|\sum e^{in\theta}|^2$; for $N \times N$ CUE matrices this average is $\min(n, N)$.

Proposition 7.1. *For $1 \leq n \leq \ell$,*

$$F(n) = \frac{q^\ell S_n - q^{2n}}{(q^\ell - 1) q^n}, \quad S_n = \sum_{\deg g = n} \Lambda(g)^2 = \sum_{e|n} e^2 I_e(q),$$

q	d	intervals	$\min \Psi/q^{d+1}$	certified	open range	time (s)
2	23	2,048	0.9961	2,048	yes	41.1
2	24	2,048	0.9964	2,048	yes	87.7
2	25	4,096	0.9983	4,096	yes	184.6
3	14	1,594,323	0.9958	1,594,323	yes	7.6
3	15	4,782,969	0.9971	4,782,969	yes	25.1
3	16	14,348,907	0.9982	14,348,907	yes	81.7
4	11	1,024	0.9981	1,024	yes	2.6
4	12	1,024	0.9968	1,024	yes	11.8
4	13	4,096	0.9995	4,096	yes	53.7
5	9	390,625	0.9961	390,625	yes	1.1
5	10	1,953,125	0.9977	1,953,125	yes	5.8
5	11	9,765,625	0.9991	9,765,625	yes	34.1
7	8	823,543	0.9982	823,543	no	3.4
7	9	5,764,801	0.9993	5,764,801	yes	26.7
7	10	40,353,607	0.9997	40,353,607	yes	212.4
8	8	512	0.9963	512	no	2.5
8	9	4,096	0.9993	4,096	no	34.1
8	10	4,096	0.9999	4,096	yes	338.6
9	6	59,049	0.9889	59,049	no	0.1
9	7	531,441	0.9987	531,441	no	0.6
9	8	4,782,969	0.9994	4,782,969	no	6.0

Table 1: Exact verification. “Intervals” is the number of Legendre classes; the ratio is $\min_f \Psi(f)/q^{d+1}$; times are wall-clock for the exact computation (two primes). Only the largest d for each q and the cases with $q \leq d - 2$ near the boundary are shown; all cases are in `data/exact_scan.csv`.

where $I_e(q)$ is the number of monic irreducible polynomials of degree e . In particular $F(n) = n - \sum_{e|n, e < n} O(q^{e-n})$, and $F(2) = 2 - 1/q + O(q^{2-\ell})$.

Proof. By Parseval, $\sum_{\chi} |\psi_n(\chi)|^2 = q^\ell \sum_x (\sum_{\deg g=n, [g]=x} \Lambda(g))^2$. For $n \leq \ell$ each class contains at most one monic polynomial of degree n , so the inner sum has at most one term and the total is $q^\ell S_n$. Remove the trivial character, $|\psi_n(1)|^2 = q^{2n}$. $\square$

The computed $F(n)$ agree with Proposition 7.1 to 10^{-14} . For $n > \ell$ there is no such formula; there $F(n)$ is the variance of prime counts in classes of degree- n polynomials, and it stays within 9% of the CUE value $\ell - 1$ (Figure 3). The nearest-neighbour spacings of the eigenangles agree with CUE($d - 2$) to within 0.05 in density.

q	d	characters	generic	roots off $ \omega = \sqrt{q}$
2	20	400,000	0.5009	0
3	12	177,146	0.6667	0
3	14	400,000	0.6672	0
4	10	262,143	0.7500	2
5	9	390,624	0.8000	0
7	8	400,000	0.8576	0

Table 2: Zeros of all Hayes L -functions for the listed (q, d) . “Generic” is the proportion of characters whose L -function has the maximal number $d - 2$ of inverse roots.

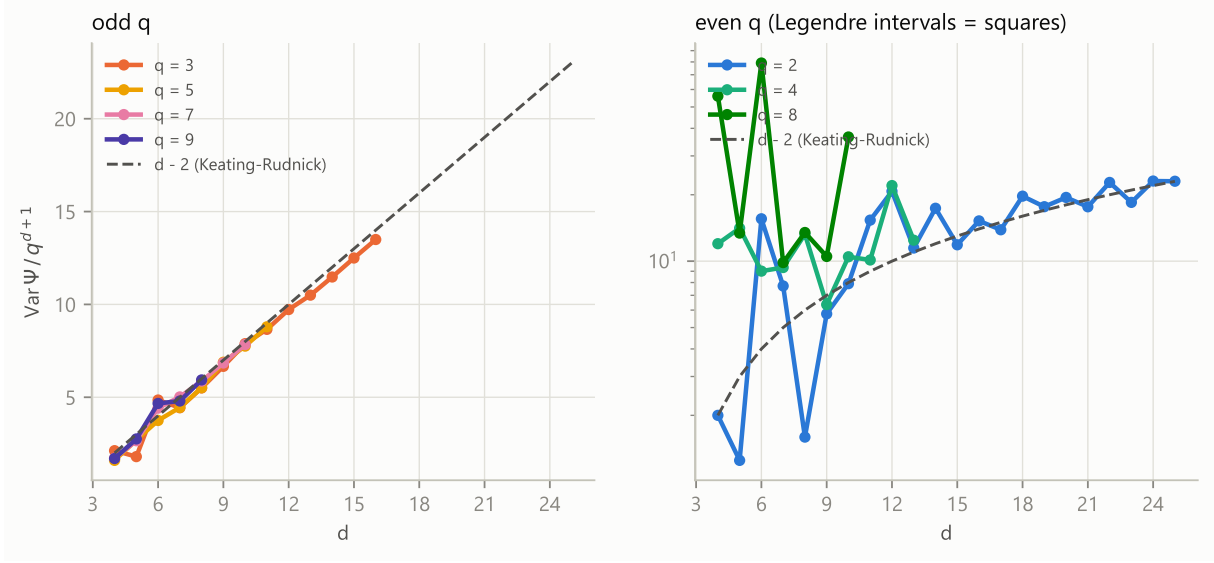


Figure 1: $\text{Var}_f \Psi(f)/q^{d+1}$ (mean square deviation from q^{d+1}) over all Legendre intervals. Left: odd q , against the Keating–Rudnick limit $d - 2$. Right: even q , where the prediction fails.

8 Characteristic 2

8.1 Structure

Let q be even. By Lemma 2.1 the Legendre intervals form the subgroup $\mathcal{G}^2$ of index $2^{k \cdot \#\{j \leq \ell \text{ odd}\}}$. Their elements have a special form.

Proposition 8.1. *Let q be even and f monic of degree d . Every $P \in \mathcal{I}_f$ can be written uniquely as*

$$P = h^2 + tb^2, \quad h = f + a, \quad \deg a \leq \lfloor d/2 \rfloor, \quad \deg b \leq \lfloor (d-1)/2 \rfloor.$$

Then $P' = b^2$, and P is squarefree if and only if $\gcd(h, b) = 1$.

Proof. Write $s = P - f^2$ as $a(t)^2 + tb(t)^2$ by separating even and odd powers (every element of $\mathbb{F}_q$ is a square); then $f^2 + s = (f + a)^2 + tb^2$ because squaring is additive. Differentiating, $P' = b^2$. If $\pi^2 \mid P$ for an irreducible π , then $\pi \mid P' = b^2$, so $\pi \mid b$ and $\pi \mid h^2$, so $\pi \mid h$. Conversely if $\pi \mid h$ and $\pi \mid b$ then π^2 divides both h^2 and tb^2 . If $b = 0$ then $P = h^2$ is not squarefree, and $\gcd(h, 0) = h$ is nonconstant. $\square$

Thus a Legendre interval in characteristic 2 is a union, over the $q^{\lfloor (d+1)/2 \rfloor}$ values of b , of the values of the quadratic polynomial $X^2 + tb^2$ on a short interval of X around f . Counting its primes is closer to a Bateman–Horn problem for quadratic polynomials than to counting primes in an interval.

8.2 The anomaly

Table 3 compares the variance of Ψ over all classes of $\mathcal{G}$ with the variance over the Legendre intervals $\mathcal{G}^2$. Over all classes the variance follows $d-2$ for even q just as for odd q , and the form factor over all characters matches CUE for $q = 2, 4, 8$ (Figure 3). Over $\mathcal{G}^2$ the variance differs from it by a factor that ranges from about 0.33 to 21, while the mean stays close to q^{d+1} . The anomaly is therefore a property of the subfamily $\mathcal{G}^2$, not of the L -functions as a whole.

By Parseval on $\mathcal{G}^2$, $\text{Var}_{\mathcal{G}^2} \Psi$ is governed by the sums $\sum_{\varepsilon} \psi_{2d}(\chi \varepsilon)$ over characters ε trivial on $\mathcal{G}^2$, i.e. the characters of order ≤ 2 . If $\psi_{2d}(\chi)$ and $\psi_{2d}(\chi \varepsilon)$ were uncorrelated, the variance would

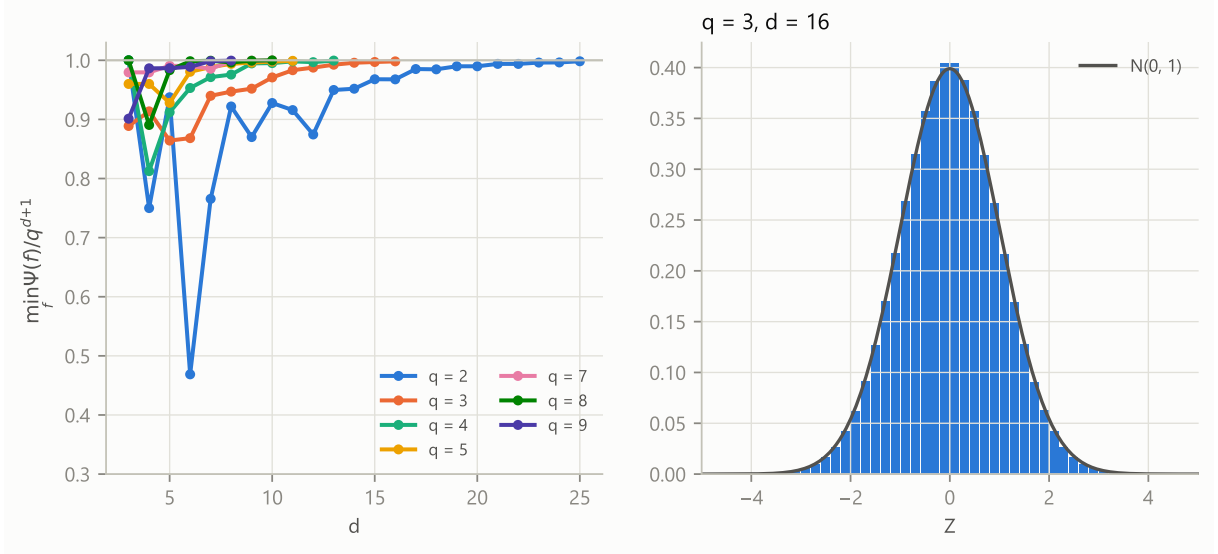


Figure 2: Left: $\min_f \Psi(f)/q^{d+1}$ against d . Right: the distribution of $Z = (\Psi - q^{d+1})/\sqrt{(d-2)q^{d+1}}$ over the 14,348,907 intervals with $(q, d) = (3, 16)$, against the standard normal density.

equal the full-group value. Define

$$\rho(\varepsilon) = \frac{\operatorname{Re} \sum_{\chi \neq 1} \psi_{2d}(\chi) \overline{\psi_{2d}(\chi \varepsilon)}}{\sum_{\chi \neq 1} |\psi_{2d}(\chi)|^2}. \quad (3)$$

The ratio of the two variances is $1 + \sum_{\varepsilon \neq 1} \rho(\varepsilon)$ up to the trivial-character terms, and in every case of Table 3 the two columns coincide to four decimals. The whole anomaly is thus carried by correlations between quadratic twists. The large correlations concentrate on quadratic characters supported on the coordinates $j > \ell/2$ (generators of order 2), and are equal across the k basis directions of $\mathbb{F}_q$; for example, for $(q, d) = (8, 4)$ every such ε has $\rho(\varepsilon) = 0.238$.

Two natural explanations fail. On fixed- b slices, $\mu(h^2 + tb^2)$ is neither of the form $\pm(-1)^{L(a)}$ with L affine over $\mathbb{F}_2$ in the coefficients of a , nor a function of $h \bmod b$ alone (tested for $d = 8, 10, 12$, $q = 2$).

Open problem 8.2. *Explain the excess variance of Ψ over Legendre intervals in characteristic 2: determine $\lim \operatorname{Var}_{G_2} \Psi/q^{d+1}$, as $q \rightarrow \infty$ through powers of 2 and as $d \rightarrow \infty$, and identify the mechanism behind the correlations $\rho(\varepsilon)$, presumably through the Berlekamp discriminant and the form $P = h^2 + tb^2$ with $P' = b^2$ [1, 8].*

9 Discussion

The explicit formula reduces the verification of the function field Legendre conjecture to a transform over the Hayes group, and exact modular arithmetic makes the result a proof. The method reaches every q for $d \leq 10$; going further needs memory proportional to q^{d-1} , so the next step, $q = 9$ with $d = 11$, needs a distributed transform.

The data support the conjecture of [2] that no Legendre interval is ever empty. For odd q the fluctuations behave as random matrix theory predicts already at $q = 3$; in characteristic 2 they do not, for a reason that Problem 8.2 asks to explain.

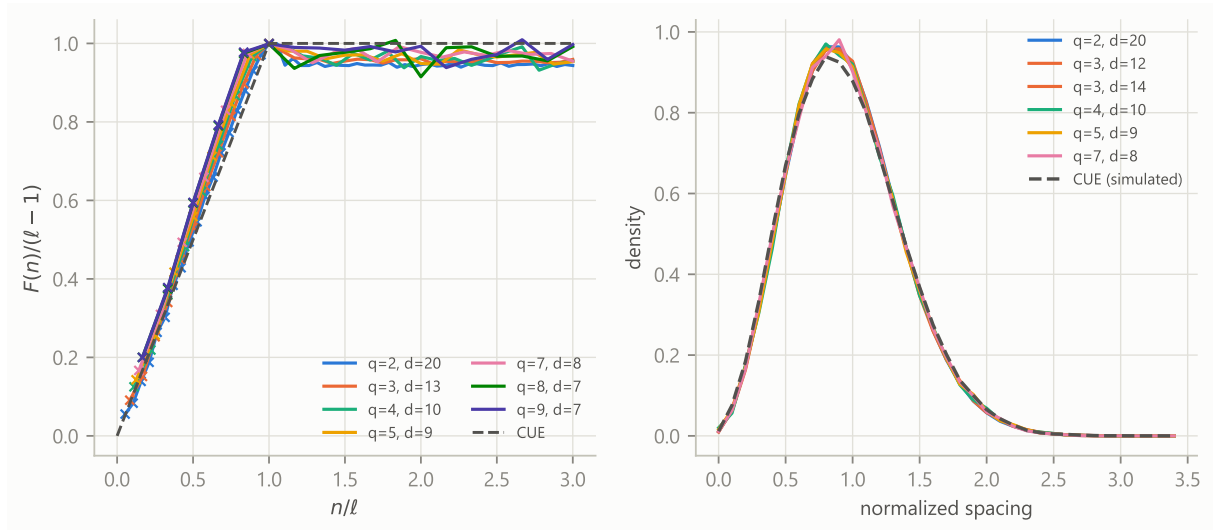


Figure 3: Left: the form factor $F(n)/(\ell - 1)$ against n/ℓ for several (q, d) , with the CUE value $\min(n, \ell - 1)/(\ell - 1)$ and the exact diagonal of Proposition 7.1 (crosses). Right: nearest-neighbour spacing densities against CUE.

References

- [1] D. Carmon, The autocorrelation of the Möbius function and Chowla’s conjecture for the rational function field in characteristic 2, *Philos. Trans. Roy. Soc. A* **373** (2015). arXiv:1409.3694.
- [2] R. Chen, Legendre’s conjecture in function fields: the classical range, full monodromy, and the open range $q \leq d - 2$ (v7), Zenodo, 2026. <https://doi.org/10.5281/zenodo.23185896>
- [3] S.D. Cohen, Explicit theorems on generator polynomials, *Finite Fields Appl.* **11** (2005), 337–357.
- [4] Z. Gao, Improved error bounds for the number of irreducible polynomials and self-reciprocal irreducible monic polynomials with prescribed coefficients over a finite field, arXiv:2109.14154.
- [5] D.R. Hayes, The distribution of irreducibles in $GF[q, x]$, *Trans. Amer. Math. Soc.* **117** (1965), 101–127.
- [6] C. N. Hsu, The distribution of irreducible polynomials in $\mathbb{F}_q[t]$, *J. Number Theory* **61** (1996), 85–96.
- [7] J. P. Keating, Z. Rudnick, The variance of the number of prime polynomials in short intervals and in residue classes, *Int. Math. Res. Not.* 2014, 259–288.
- [8] W. Sawin, M. Shusterman, On the Chowla and twin primes conjectures over $\mathbb{F}_q[T]$, *Ann. of Math.* **196** (2022). arXiv:1808.04001.

q	d	mean	Var (all classes)	Var (Legendre)	ratio	excess	$d - 2$
2	4	0.7500	1.50	2.00	1.33	1.33	2
2	5	1.0625	1.17	1.25	1.07	1.07	3
2	6	1.0625	5.76	15.56	2.70	2.70	4
2	7	1.0469	3.88	7.72	1.99	1.99	5
2	8	0.9688	4.88	1.59	0.33	0.33	6
2	9	0.9805	7.10	5.77	0.81	0.81	7
2	10	1.0273	7.19	7.91	1.10	1.10	8
2	11	0.9985	8.57	15.35	1.79	1.79	9
2	12	0.9839	9.75	20.74	2.13	2.13	10
2	13	1.0044	10.10	11.47	1.14	1.14	11
2	14	0.9995	11.30	17.37	1.54	1.54	12
2	15	0.9995	12.10	11.86	0.98	0.98	13
2	16	0.9983	13.19	15.22	1.15	1.15	14
4	4	1.0000	2.25	12.00	5.33	5.33	2
4	5	1.0117	2.60	14.11	5.43	5.43	3
4	6	0.9883	3.70	9.01	2.44	2.44	4
4	7	1.0037	5.23	9.36	1.79	1.79	5
4	8	1.0000	5.88	13.25	2.25	2.25	6
4	9	0.9999	6.79	6.36	0.94	0.94	7
8	4	0.9727	2.62	56.00	21.33	21.33	2
8	5	1.0017	3.11	13.41	4.31	4.31	3
8	6	1.0017	4.75	79.09	16.66	16.66	4
8	7	1.0002	4.96	9.85	1.99	1.99	5

Table 3: Characteristic 2. The two variance columns are the mean of $(\Psi - q^{d+1})^2/q^{d+1}$ over all classes of $\mathcal{G}$ and over the Legendre intervals $\mathcal{G}^2$; “mean” is the mean of Ψ/q^{d+1} over $\mathcal{G}^2$; “ratio” is the quotient of the two variances, and “excess” is $1 + \sum_{\varepsilon \neq 1} \rho(\varepsilon)$ from (3). Source: `data/char2_variance.csv`.